

Assignment-based Subjective Questions

1. From your analysis of the categorical variables from the dataset, what could you infer about their effect on the dependent variable?

Ans. In the bike-sharing dataset, let's consider the effect of the categorical variable 'weathers' on the target variable 'count'. While performing EDA, I visualized the relationship between the categorical variables and the target variable. It was seen that during the weathersit_3 (Light Snow, Light Rain + Thunderstorm + Scattered clouds, Light Rain + Scattered), decreases in the bike hires numbers by 0.333164 units have been seen.

approximately. Similarly, certain inferences could be made by season_Spring whereas season_Winter shows reverse trend.

Also, during model building on the inclusion of categorical features such as pre season etc we saw a significant change in the value of R-squared and adjusted R-squared. This implies that the categorical features helped explain a greater proportion of variances in the data sets.

2. Why is it important to use drop_first=True during dummy variable creation?

Ans. During dummy value creation (dummy encoding) it is advisable to use **drop_first=True**, otherwise we will get a redundant feature i.e. dummy variables might be correlated because the first column becomes a reference group during dummy encoding.

3. Looking at the pair-plot among the numerical variables, which one has the highest correlation with the target variable?

Ans. The numerical variable 'registered (0.95)' has the highest correlation with the target variable 'cnt', if we consider all the features. But after data preparation, when we drop registered due to multi collinearity the numerical variable 'atemp (0.63)' has the highest correlation with the target variable 'cnt'.

4. How did you validate the assumptions of Linear Regression after building the model on the training set?

Ans. After building the model on the training set, I carried out the following analysis: -

Assumptions of Linear Regression:

1. There is a linear relationship between X and Y

2. Error terms are normally distributed with mean zero (not X, Y)
3. Residual Analysis of Training Data proves that the Residuals are normally distributed.
4. Hence our assumption for Linear Regression is valid.
5. Eliminations and inclusion of independent variables into each model based on VIF and p-values to avoid multi collinearity.

5. Based on the final model, which are the top 3 features contributing significantly towards explaining the demand of the shared bikes?

Ans. As per our final Model, the top 3 predictor variables that influences the bike booking are:

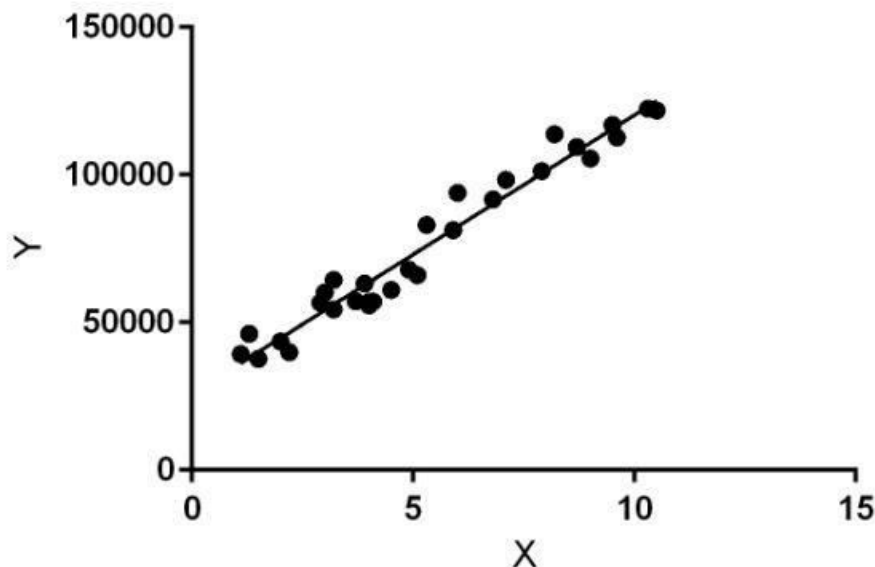
1. **Temperature (temp)** - A coefficient value of '0.375922' indicated that a unit increase in temp variable increases the bike hire numbers by 0.375922 units.
2. **Weather Situation 3 (weathersit_3)** (Light Snow, Light Rain + Thunderstorm + Scattered clouds, Light Rain + Scattered) - A coefficient value of '-0.333164' indicated that, w.r.t Weathersit_3, a unit increase in Weathersit_3 variable decreases the bike hire numbers by 0.333164 units.
3. **Year (yr)** - A coefficient value of '0.232965' indicated that a unit increase in yr variable increases the bike hires numbers by 0.232965 units.

So, it's suggested to consider these variables utmost importance while planning, to achieve maximum Booking.

General Subjective Questions

1. Explain the linear regression algorithm in detail.

Ans. Linear Regression is a machine learning algorithm based on supervised learning. It performs a regression task. Regression models a target prediction value based on independent variables. It is mostly used for finding out the relationship between variables and forecasting. Different regression models differ based on – the kind of relationship between dependent and independent variables, they are considering and the number of independent variables being used.



Linear regression performs the task to predict a dependent variable value (y) based on a given independent variable (x). So, this regression technique finds out a linear relationship between x (input) and y(output). Hence, the name is Linear Regression.

The linear regression model can be represented by the following equation:

$$Y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_n x_n$$

where,

Y is the predicted value

θ_0 is the constant term.

$\theta_1, \dots, \theta_n$ are the model parameters

$x_1, x_2, \dots, x_n$ are the feature values.

The goal of regression analysis is to create a trend line based on the data you have gathered. This then allows you to determine whether other factors apart from the amount of calories consumed affect your weight, such as the number of hours you sleep, work pressure, level of stress, type of exercises you do etc. Before taking into account, we need to look at these factors and attributes and determine whether there is a correlation between them. Linear Regression can then be used to draw a trend line which can then be used to confirm or deny the relationship between attributes. If the test is done over a long time duration, extensive data can be collected and the result can be evaluated more accurately.

How to update θ_1 and θ_2 values to get the best fit line?

Cost Function (J): By achieving the best-fit regression line, the model aims to predict y value such that the error difference between predicted value and true value is minimum. So, it is very important to update the θ_1 and θ_2 values, to reach the best value that minimize the error between predicted y value (pred) and true y value (y).

$$\text{minimize } \frac{1}{n} \sum_{i=1}^n (\text{pred}_i - y_i)^2$$

$$J = \frac{1}{n} \sum_{i=1}^n (\text{pred}_i - y_i)^2$$

Cost function(J) of Linear Regression is the **Root Mean Squared Error (RMSE)** between predicted y value (pred) and true y value (y).

Gradient Descent: To update θ_1 and θ_2 values in order to reduce Cost function (minimizing RMSE value) and achieving the best fit line the model uses Gradient Descent. The idea is to start with random θ_1 and θ_2 values and then iteratively updating the values, reaching minimum cost

2. Explain the Anscombe's quartet in detail.

Ans. It is a group of four datasets that appear to be similar when using typical summary statistics, yet tell four different stories when graphed

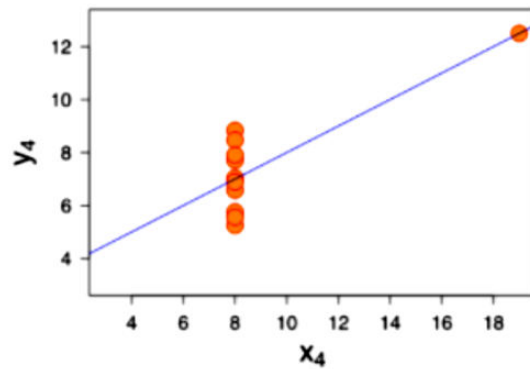
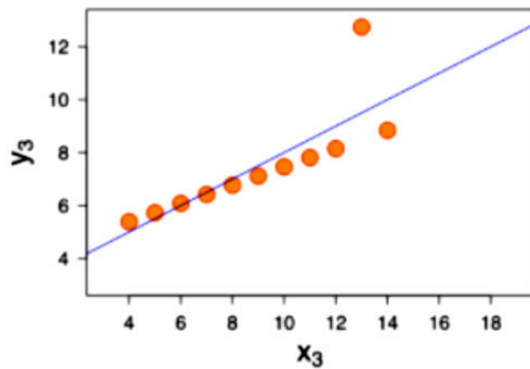
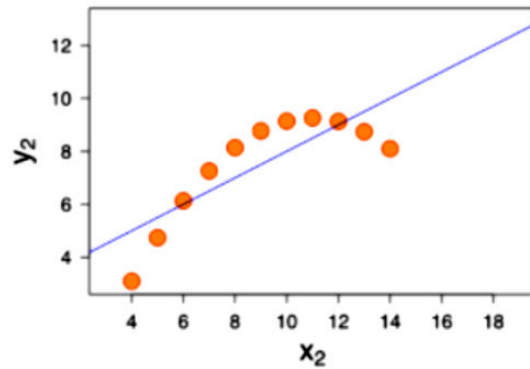
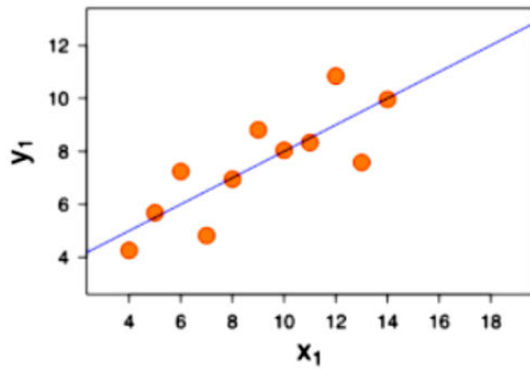
All the summary statistics for each dataset are identical

1. The average value of x is 9.
2. The average value of y is 7.5.
3. The variance for x is 11 and y is 4.12
4. The correlation between x and y is 0.816
5. The line of best fit is $y = 0.5x + 3$.

But the plots tell a different and unique story for each dataset.

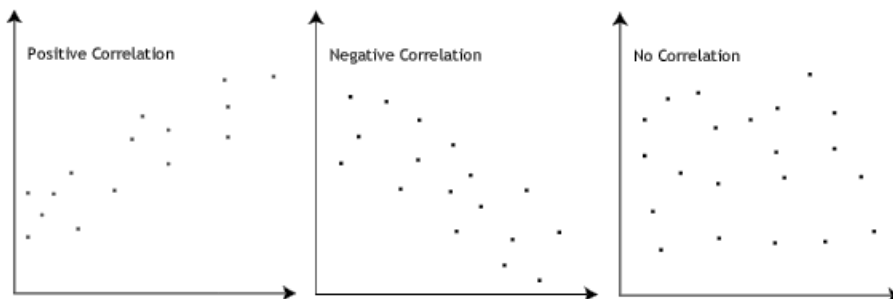
☐ The first scatter plot (top left) appears to be a simple linear relationship, corresponding to two variables correlated where y could be modelled as gaussian with mean linearly dependent on x.

☐ The second graph (top right) is not distributed normally; while a relationship between the two variables is obvious, it is not linear, and the Pearson correlation coefficient is not relevant. A more general regression and the corresponding coefficient of determination would be more appropriate.



In the third graph (bottom left), the distribution is linear, but should have a different regression line (a robust regression would have been called for). The calculated regression is offset by the one outlier which exerts enough influence to lower the correlation coefficient from 1 to 0.816.

Finally, the fourth graph (bottom right) shows an example when one high-leverage point is enough to produce a high correlation coefficient, even though the other data points do not indicate any relationship between the variables.



3. What is Pearson's R?

Ans. Pearson's R is a numerical summary of the strength of the linear association between the variables. It varies between -1 and +1. If the variables tend to go up and down together, the correlation coefficient will be positive. If the variables tend to go up and down in opposition with low values of one variable associated with high values of the other, the correlation coefficient will be negative. $r = 1$ means the data is perfectly linear with a positive slope (i.e., both variables tend to change in the same direction)

$r = -1$ means the data is perfectly linear with a negative slope (i.e., both variables tend to change in different directions)

$r = 0$ means there is no linear association

$r > 0 < 5$ means there is a weak association

$r > 5 < 8$ means there is a moderate association

$r > 8$ means there is a strong association

$$r = \frac{N\sum xy - (\sum x)(\sum y)}{\sqrt{[N\sum x^2 - (\sum x)^2][N\sum y^2 - (\sum y)^2]}}$$

Where:

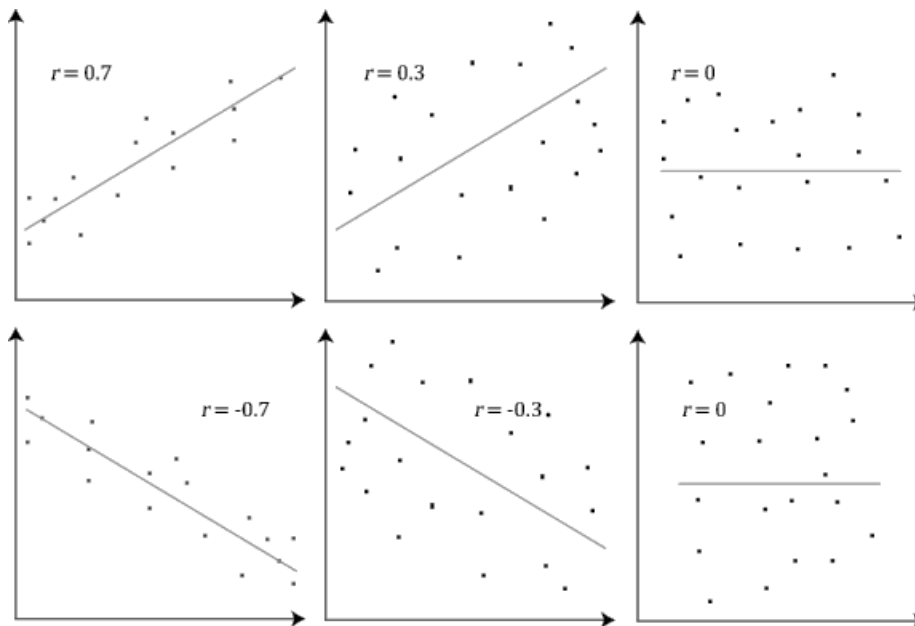
N	=	number of pairs of scores
$\sum xy$	=	sum of the products of paired scores
$\sum x$	=	sum of x scores
$\sum y$	=	sum of y scores
$\sum x^2$	=	sum of squared x scores
$\sum y^2$	=	sum of squared y scores

The Pearson product-moment correlation coefficient (or Pearson correlation coefficient, for short) is a measure of the strength of a linear association between two variables and is denoted by r .

Basically, a Pearson product-moment correlation attempts to draw a line of best fit through the data of two variables, and the Pearson correlation coefficient, r , indicates how far away all these data points are to this line of best fit (i.e., how well the data points fit this new model/line of best fit).

The Pearson correlation coefficient, r , can take a range of values from +1 to -1. A value of 0 indicates that there is no association between the two variables. A value greater than 0 indicates a positive association; that is, as the value of one variable increases, so does the value of the other variable. A value less than 0 indicates a negative association; that is, as the value of one variable increases, the value of the other variable decreases.

The stronger the association of the two variables, the closer the Pearson correlation coefficient, r , will be to either +1 or -1 depending on whether the relationship is positive or negative, respectively. Achieving a value of +1 or -1 means that all your data points are included on the line of best fit – there are no data points that show any variation away from this line. Values for r between +1 and -1 (for example, $r = 0.8$ or -0.4) indicate that there is variation around the line of best fit. The closer the value of r to 0 the greater the variation around the line of best fit



4. What is scaling? Why is scaling performed? What is the difference between normalized scaling and standardized scaling?

Ans. Scaling is a technique to standardize the independent features present in the data in a fixed range. It is performed during the data pre-processing to handle highly varying magnitudes or values or units. It is extremely important to rescale the variables so that they have a comparable scale. If we don't have comparable scales, then some of the coefficients as obtained by fitting the regression model might be very large or very small as compared to the other coefficients.

Normalized scaling means to scale a variable to have values between 0 and 1, while standardized scaling refers to transform data to have a mean of zero and a standard deviation of 1

What?

It is a step of data Pre-Processing which is applied to independent variables to normalize the data within a particular range. It also helps in speeding up the calculations in an algorithm.

Why?

Most of the times, collected data set contains features highly varying in magnitudes, units and range. If scaling is not done, then algorithm only takes magnitude in account and not units hence incorrect modelling. To solve this issue, we have to do scaling to bring all the variables to the same level of magnitude.

It is important to note that scaling just affects the coefficients and none of the other parameters like t-statistic, F-statistic, p-values, R-squared, etc.

Normalization/Min-Max Scaling:

☐ It brings all of the data in the range of 0 and 1.

$$\text{MinMax Scaling: } x = \frac{x - \min(x)}{\max(x) - \min(x)}$$

❑ `sklearn.preprocessing.MinMaxScaler` helps to implement normalization in python.

Standardization Scaling:

❑ Standardization replaces the values by their Z scores. It brings all of the data into a standard normal distribution which has mean (μ) zero and standard deviation one (σ).

$$\text{Standardisation: } x = \frac{x - \text{mean}(x)}{\text{sd}(x)}$$

❑ `sklearn.preprocessing.scale` helps to implement standardization in python.

❑ One disadvantage of normalization over standardization is that it **loses** some information in the data, especially about **outliers**.

5. You might have observed that sometimes the value of VIF is infinite. Why does this happen?

Ans. The Variance Inflation Factor (VIF) is a measure of colinearity among predictor variables within a multiple regression. It is calculated by taking the ratio of the variance of all a given model's betas divide by the variance of a single beta if it were fit alone.

If there is perfect correlation, then VIF = infinity. A large value of VIF indicates that there is a correlation between the variables. An infinite VIF value indicates that the corresponding variable may be expressed exactly by a linear combination of other variables (which show an infinite VIF as well).

6. What is a Q-Q plot? Explain the use and importance of a Q-Q plot in linear regression.

Ans. Quantile-Quantile (Q-Q) plot, is a graphical tool to help us assess if a set of data plausibly came from some theoretical distribution such as a Normal, exponential or Uniform distribution. Also, it helps to determine if two data sets come from populations with a common distribution.

